

On the Stability of Latency-Aware Cache Hierarchies

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Abstract

We examine the behaviour of consensus protocols in low-bandwidth links. The measurements reported here were collected over twelve independent trials under controlled conditions. We release the processing scripts and intermediate artefacts to support independent verification. Taken together, these findings argue for treating the two regimes separately in future analyses.

1. Introduction

Calibration drift over the trial period remained below the threshold set out in the protocol. We release the processing scripts and intermediate artefacts to support independent verification. Neither of the alternative formulations improved on the simple additive model in our setting. Taken together, these findings argue for treating the two regimes separately in future analyses. Sensitivity to the smoothing window proved modest across the range we examined. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. Residual error grows approximately linearly with the number of coupled parameters. Longer observation windows narrow the confidence interval but do not shift the central estimate.

Discrepancies at the boundary are consistent with the simplifying assumptions noted above. Taken together, these findings argue for treating the two regimes separately in future analyses. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. We release the processing scripts and intermediate artefacts to support independent verification. The reduced model retains most of the explanatory power at a fraction of the computational cost. The measurements reported here were collected over twelve independent trials under controlled conditions. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study.

2. Related Work

Neither of the alternative formulations improved on the simple additive model in our setting. Replicate agreement was high, suggesting the protocol is robust to small handling differences. A modest but repeatable bias appears in the lowest-magnitude bin and warrants further study. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity. Residual error grows approximately linearly with the number of coupled parameters. Our baseline reproduces the behaviour described in earlier work to within the stated tolerance.

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3. Materials and Methods

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4. Experimental Setup

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Discrepancies at the boundary are consistent with the simplifying assumptions noted above. The reduced model retains most of the explanatory power at a fraction of the computational cost. Residual error grows approximately linearly with the number of coupled parameters. Longer observation windows narrow the confidence interval but do not shift the central estimate. Neither of the alternative formulations improved on the simple additive model in our setting. We observe a pronounced inflection once the load exceeds roughly sixty percent of nominal capacity.

5. Results

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